EECS 484, Winter 2026- Database Management Systems

Database Application Programming

Chapter 6

JDBC Section

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Random Slide

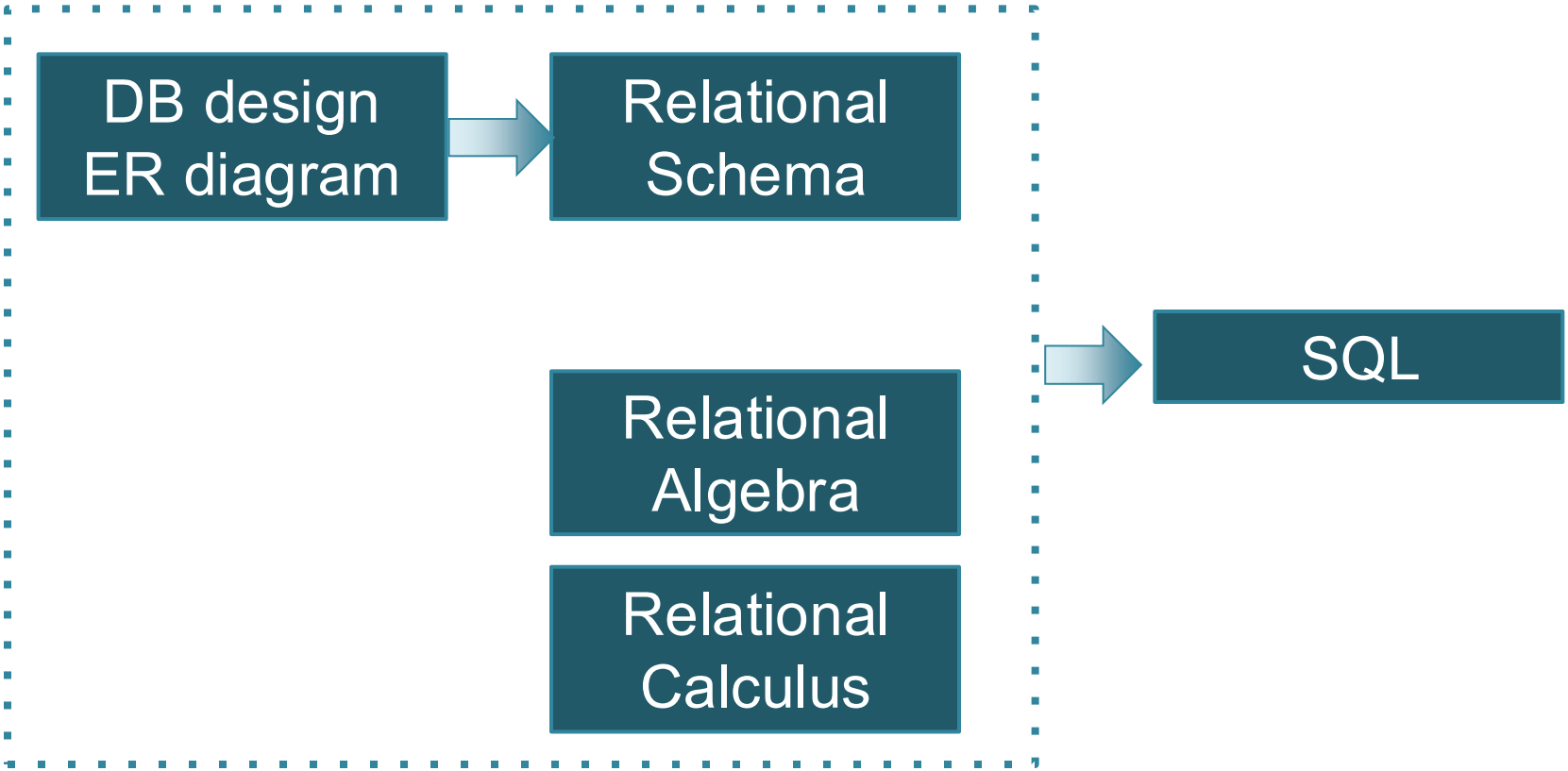
Tea Time: Karak Tea

Karak Chai is a strongly brewed tea with some spices and milk.

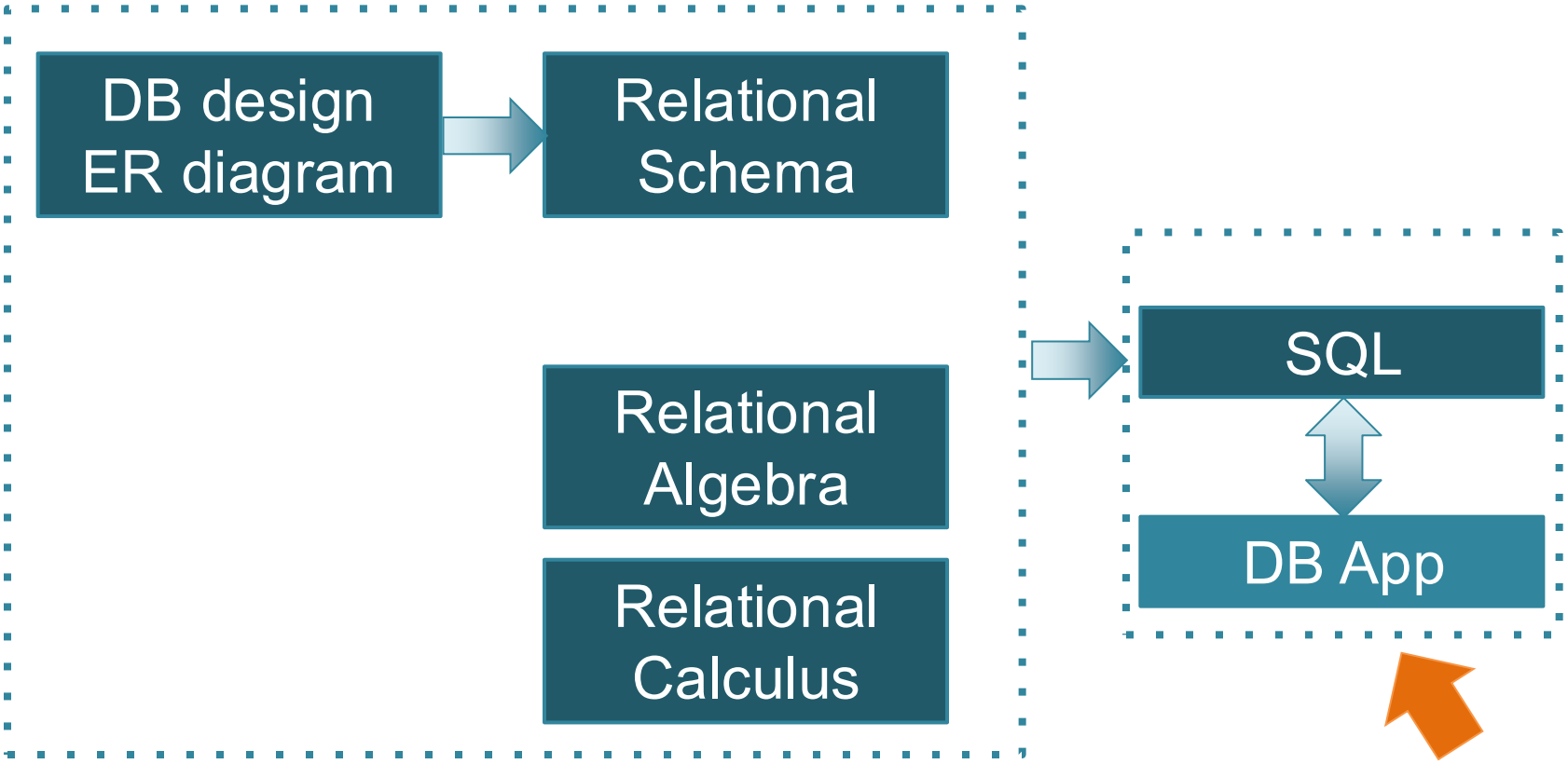
Common spices:
Cinnamon, cloves, (saffron), ginger

Origin: India/Pakistan
In Hindi the word 'Karak' means strong and 'Chai' means tea

Review


DB design
ER diagram
Relational
Schema
Relational
Algebra
Relational
Calculus
SQL

Today


DB design
ER diagram
Relational
Schema
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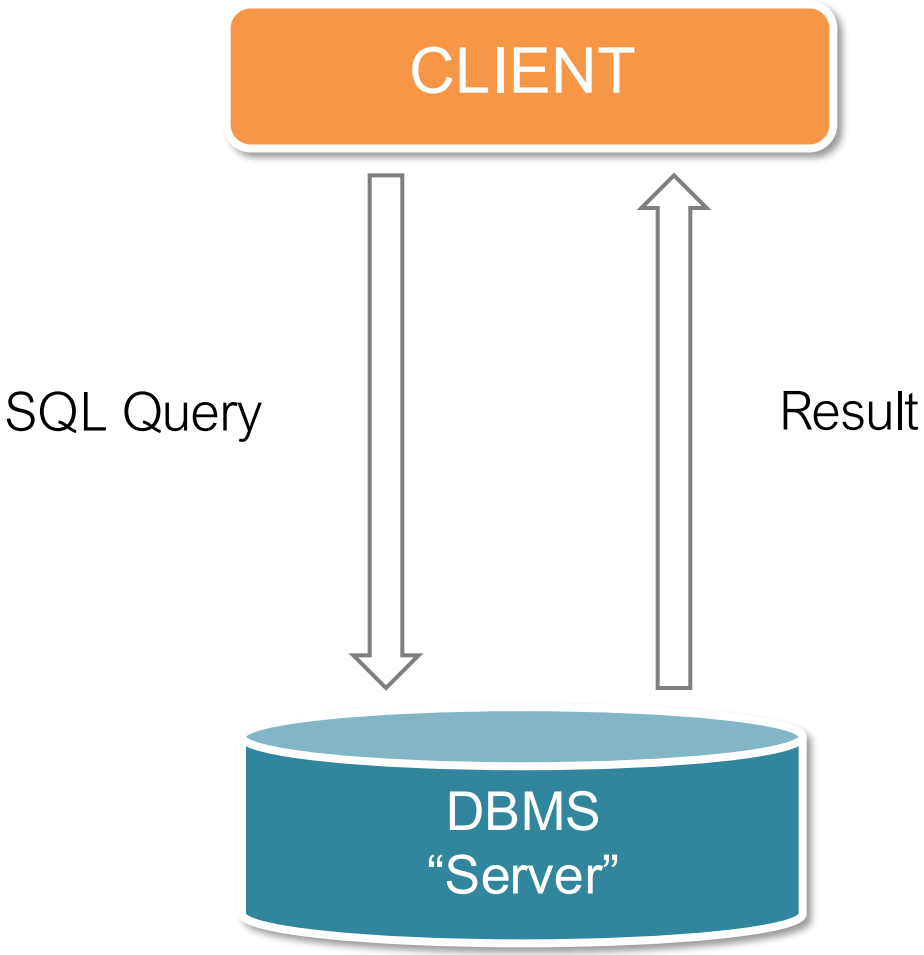
Algebra
Relational
Calculus
SQL
DB App

Databases “In the Wild”

- So far, we’ve talked about the DBMS as a standalone system
  - Access interactively by writing SQL queries (e.g., using SQL*Plus)
- In practice, DBMS is often part of a larger software infrastructure
  - Multi-tiered system architecture
  - Access database from another program

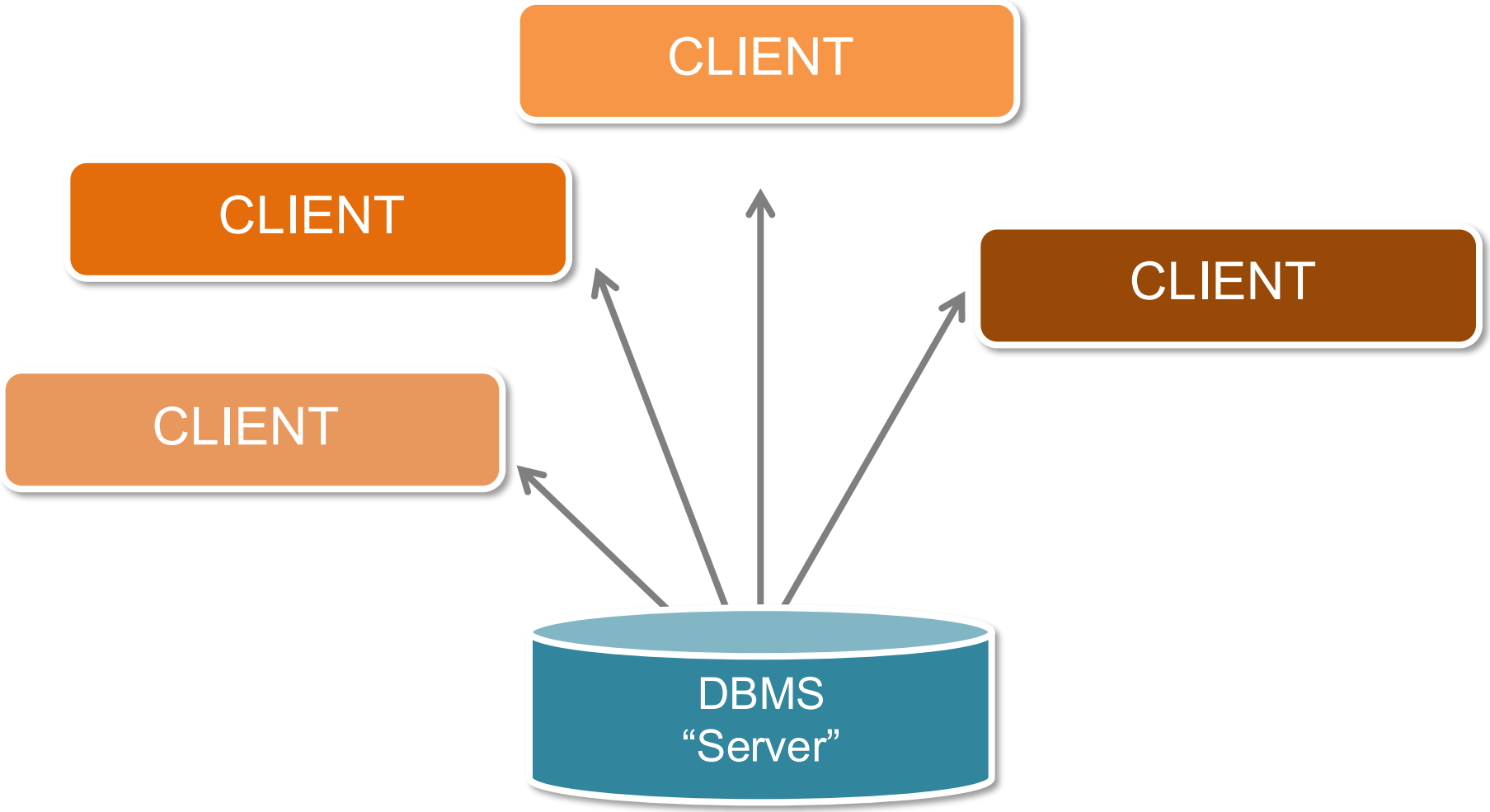
Database "Ecosystem" (1)

"Client-Server Architecture"



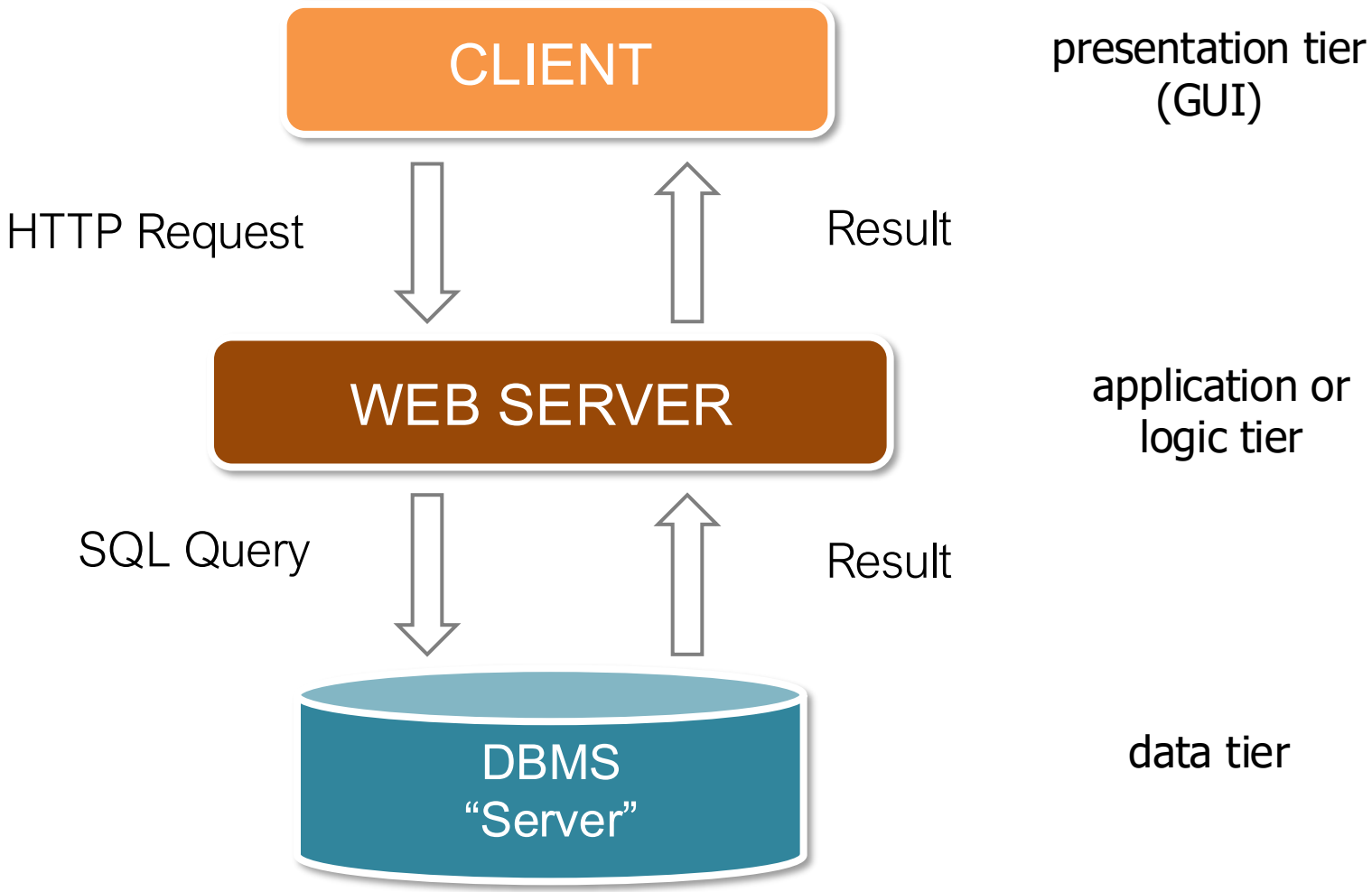
Many Clients

"Client-Server Architecture"

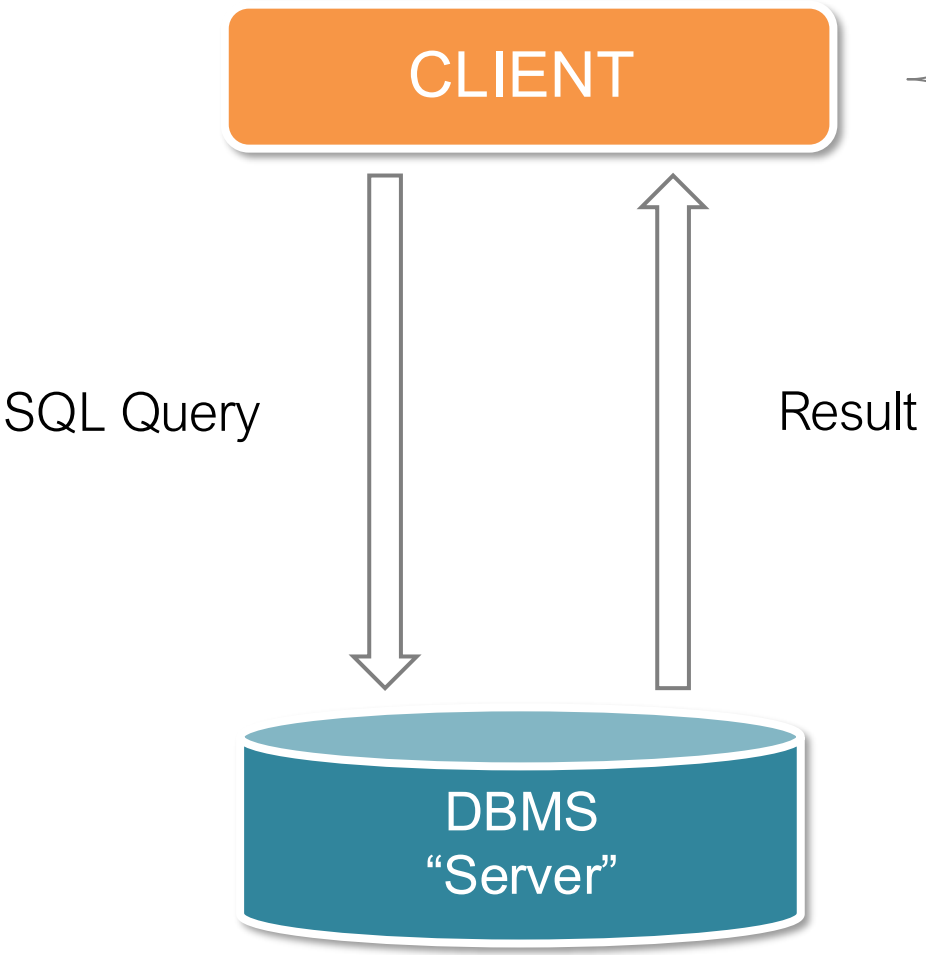


Database "Ecosystem" (2)

"3-Tier Architecture" (Common to add more tiers, too)



Embedding SQL in Application



Client is often a program, written in a language like C++ or Java

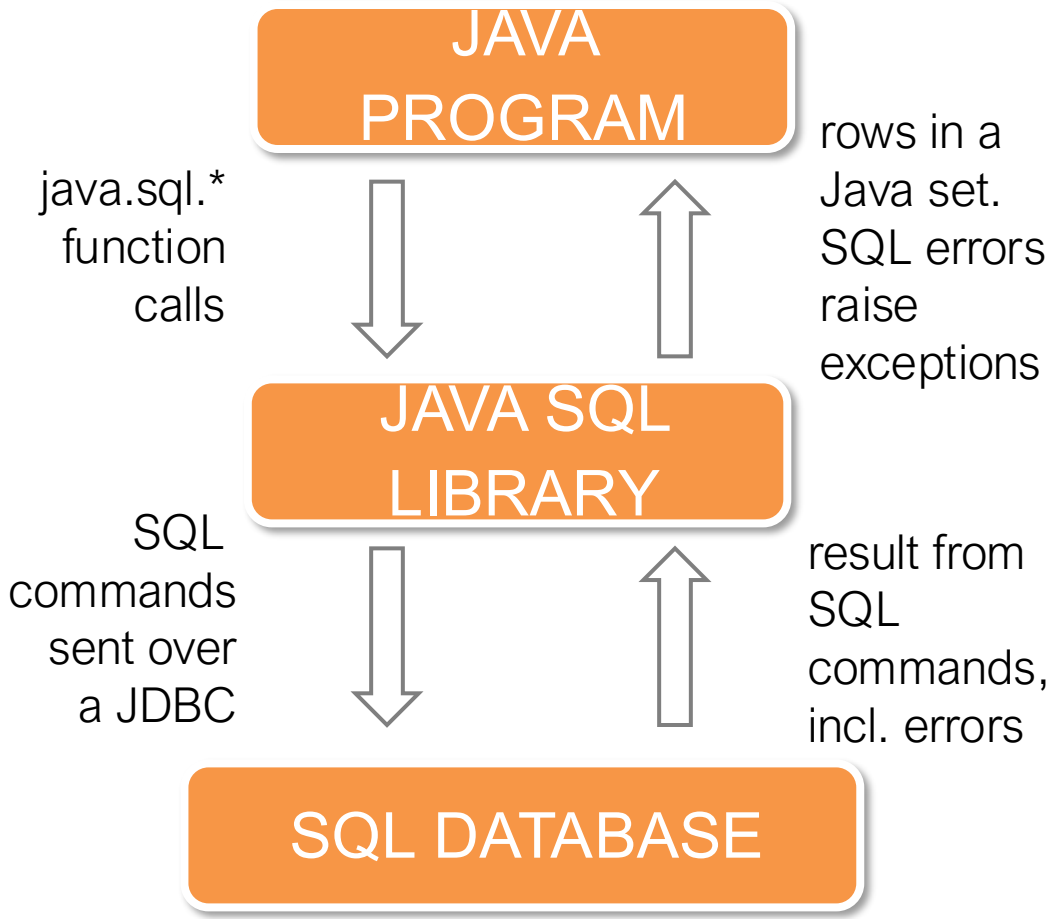
CHALLENGE:
How to access SQL from application code?

SQL Integration with PL

- Ugly Problem:
  - Database supports SQL queries, apps written in programming languages (e.g. Java, C++, Python)
  - What is the interface between the two?
- A Common Solution:
  - “Embed” SQL in host language
  - Provide an API for processing query results
- DBMS only stores scalar values (integers, strings)
  - Programmers has to convert complex objects into groups of simple values that a DB understands
  - That’s what object-relational mapping tools do
    - Examples: LINQ to SQL, Ruby on Rails…

JDBC ("Java-Database Connectivity")

- Connect to a database using a JDBC driver
- Send queries over the JDBC connection
- Receive results into a Java ResultSet



Try things out with sqlite / Oracle

- Download JDBCTest.java and the latest jdbc driver for sqlite from https://bitbucket.org/xerial/sqlite-jdbc/downloads
- Alternative: Oracle driver posted with Project 2.
- Compile: javac JDBCTest.java
- Run: java –cp .:*sqlite-jdbc-3.7.2.jar* JDBCTest

Opening/Closing Connections

- Get Connection object:
  - Oracle requires passwords
  - Sqlite3 is file-based and does not
    - Connection conn = ….
- Always close connections before quitting the program
  - conn.close();
- A cool trick in recent Java/JDBC to auto-close:

http://docs.oracle.com/javase/7/docs/technotes/guides/jdbc/jdbc_41.html

JDBC Example

```
Connection conn;
// Obtain a connection to DB, store in conn
// (Requires JDBC driver; See sample code in Project 2)

String q = "SELECT name FROM Students WHERE GPA > 3.5";
try {
     Statement st = conn.createStatement();
     ResultSet rs = st.executeQuery(q);

     while (rs.next()) {
        String name = rs.getString(”name");
        System.out.println(name);
     }
     rs.close();
     st.close();
}
catch(SQLException e){System.err.println(e.getMessage());}
```

Cursor retrieves rows from result one at a time

Full Javadoc for java.sql available online:
http://download.oracle.com/javase/6/docs/api/

Challenge 1

- DBMS and PL implement different types
  - "Impedance Mismatch"
- Need to match DB types with PL types

SQL Type	Java Type	ResultSet Method
CHAR	String	getString()
VARCHAR	String	getString()
DOUBLE	Double	getDouble()
INTEGER	Integer	getInt()
NUMBER	(depends)	(depends)

Challenge 2

- What computation to do in the database vs. the application program?
- Rules of Thumb:
  - Avoid fetching more data than necessary
  - “Push” data processing to the DBMS when possible

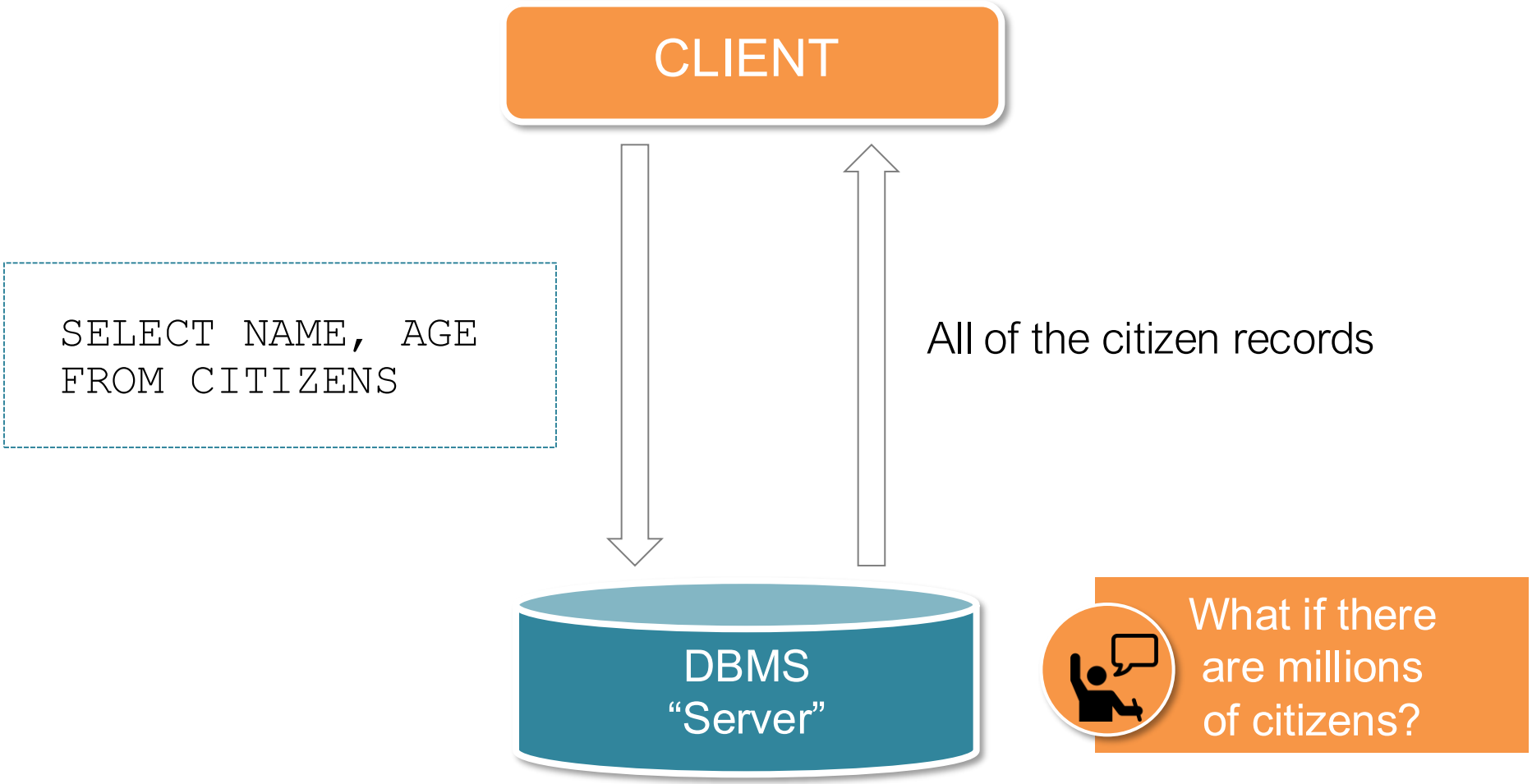
JDBC Example

What does the following code snippet do?

```
String query = "SELECT NAME, AGE FROM CITIZENS";
try {
      double sum = 0;
      double count = 0;
     Statement st = conn.createStatement();
     ResultSet rs = st.executeQuery(query);
     while (rs.next()) {
        String name = rs.getString("NAME");
        sum += rs.getDouble("AGE");
         count++;
     }
      System.out.println(sum/count);
     // be sure to close rs and st as well here
     // and in exception handlers. Omitted for brevity.
}
catch(SQLException e){System.err.println(e.getMessage());}
```

What Happens?

Compute the average age



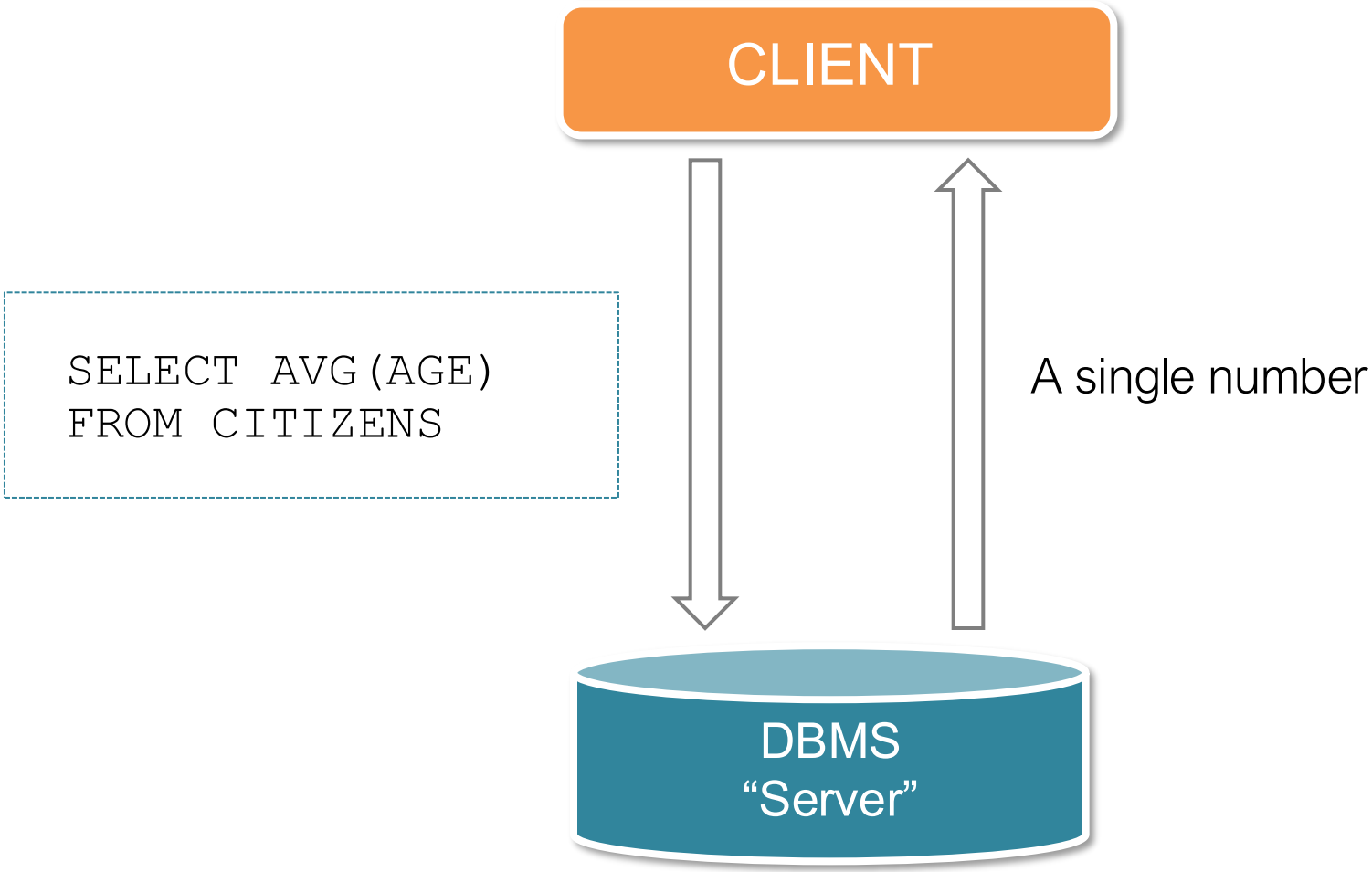
JDBC Example (Revised)

Push the computation “closer” to the data…
Make DBMS do processing it does well.

```
String query = "SELECT AVG(AGE) FROM CITIZENS";
try {
     Statement st = conn.createStatement();
     ResultSet rs = st.executeQuery(query);
     while (rs.next()) {
        Double avg = rs.getDouble(1);
         System.out.println(avg);
     }
     // close omitted for brevity
}
catch(SQLException e){System.err.println(e.getMessage());}
```

What Happens Now?

Compute the average age



Statement vs. PreparedStatement

- Statement (Base class)
  - Arbitrary SQL query
  - DBMS parses and optimizes each query
- PreparedStatement
  - Parameterized SQL query
  - DBMS “pre-compiles” the query (parses, optimizes, and stores query execution plan)
  - Can be used multiple times
  - Amortizes optimization cost across multiple uses

Statement vs. PreparedStatement

```
public List<String> getNames (int age) {
   String q = "SELECT name FROM Students WHERE age = " + age;
   Statement st = conn.createStatement();
   ResultSet rs = st.executeQuery(q);
   …
}
```

```
public List<String> getNames (int age) {
   String q = "SELECT name FROM Students WHERE age = ?";
   PreparedStatement ps = conn.prepareStatement(q);
   ps.setDouble(1, age);
   ResultSet rs = ps.executeQuery();
   …
}
```

Which is more efficient if getNames() called many times?

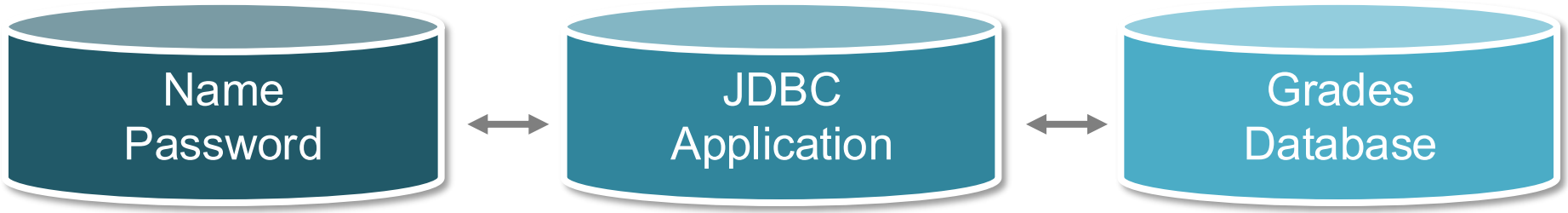
PreparedStatement

- PreparedStatement also does some type checking on parameters and removal of escape characters
- Helpful for preventing some security problems (e.g. SQL Injection)

Security Issues / SQL Injection

- Common vulnerability in database applications
  - SQL Injection is simple, yet surprisingly common
  - Can be prevented with defensive coding

Web Front-End



Students
(Name,Grade,Password)

SQL Injection – Example

JDBC

```
Authenticate (String n, String p) {
…
String query =
"SELECT grade FROM students WHERE name = " + n + "'
AND password = '" + p + "' ";
…
}
```

Input:

name = bart; password = mypword

SQL:

```
SELECT grade FROM students WHERE name = 'bart' AND
password = 'mypword'
```

SQL Injection – Example

JDBC

```
Authenticate (String n, String p) {
…
String query =
"SELECT grade FROM students WHERE name =
‘" + n + " ’  AND password = ‘" + p + "’ ";
…
}
```

Input:

name = lisa; password = n’ OR ‘x’=‘x

SQL:

```
SELECT grade FROM students WHERE name = ‘lisa’ AND
password = ‘n’ OR ‘x’ = ‘x’
```

SQL Injection – Example

JDBC

```
Authenticate (String n, String p) {
…
String query =
"SELECT grade FROM students WHERE name =
‘" + n + " ’  AND password = ‘" + p + "’ ";
…
}
```

Input:

```
name= foo; password= n’; UPDATE students SET grade= ‘A
```

SQL:

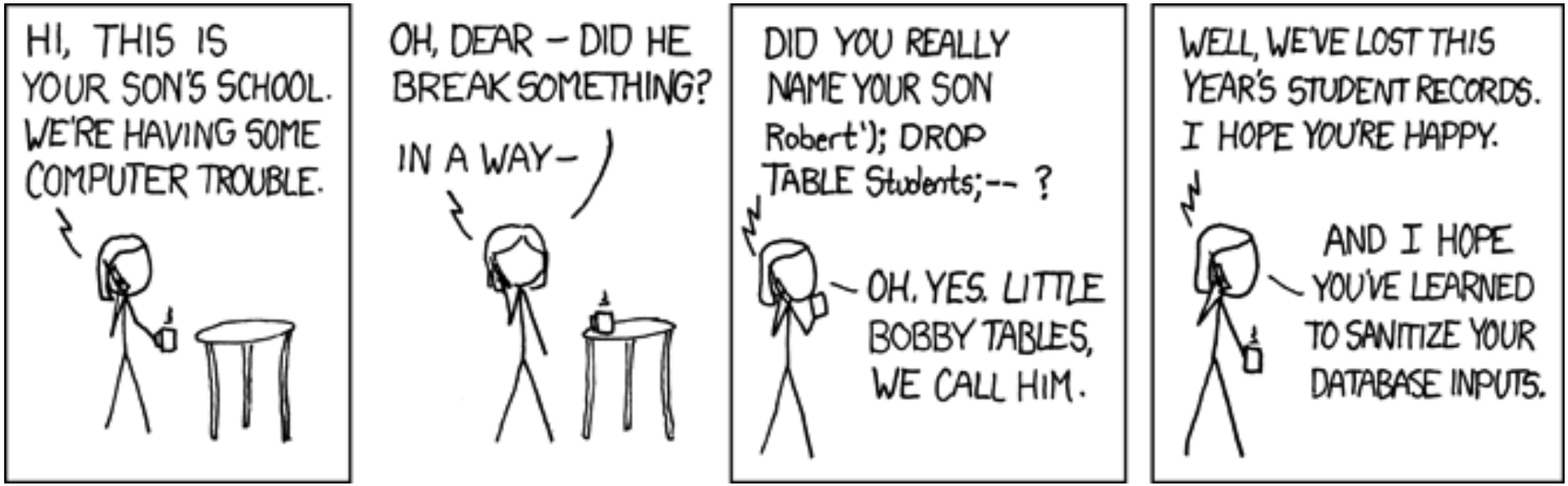
```
SELECT grade FROM students WHERE name = ‘foo’ AND
 password = ‘n’; UPDATE students SET grade = ‘A’
```

SQL Injection

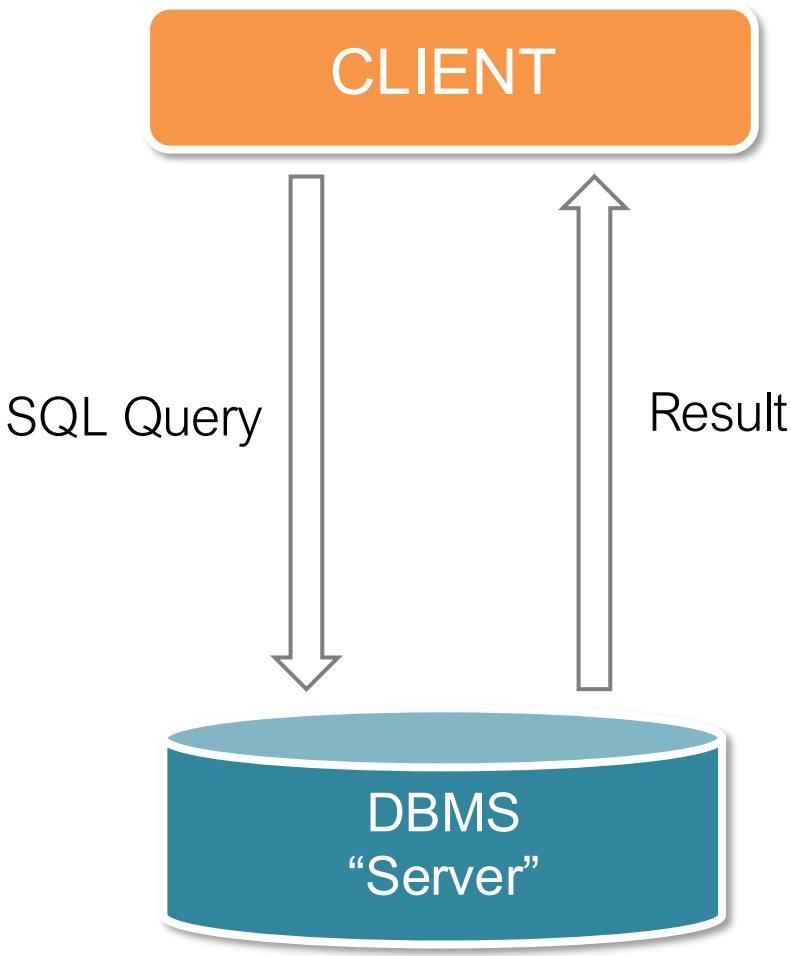
What is a simple way to prevent this kind of attack?

Validate input!

Use a prepared statement for user input parameters!



Stored Procedures and UDFs



Database applications you write (e.g., using JDBC) are usually located outside the DBMS.

Exceptions:

- Stored Procedures
- User-Defined Functions (UDFs)
- Stored and executed within the DBMS

Syntax is optional

Example of a Stored Procedure

- Define using “CREATE PROCEDURE”, call using “Call”
  - Every DB vendor has slightly diff syntax, below is for MySQL

```
DELIMITER //

CREATE PROCEDURE give_raise(IN p_emp_id INT, IN p_amount DECIMAL(10,2))
BEGIN
  UPDATE employees
  SET salary = salary + p_amount
  WHERE emp_id = p_emp_id;
END//

DELIMITER ;

CALL give_raise(42, 1000.00);
```

Purpose of a Stored Proc: perform **actions / workflow**, may return results via OUT params or result sets

Syntax is optional

Example of a User Defined Func.

- Define using “CREATE FUNCTION”, you can use it in SELECT, WHERE, JOIN condition, anywhere you can use a function
  - Every DB vendor has slightly diff syntax, below is for MySQL

```
DELIMITER //

CREATE FUNCTION add_tax(p_price DECIMAL(10,2))
RETURNS DECIMAL(10,2)
DETERMINISTIC
RETURN p_price * 1.06//

DELIMITER ;

SELECT add_tax(100.00);
```

Purpose of a UDF: compute and **return a value** (scalar or table, depending on DBMS)

When would you use DB design?

Anytime!

Summary

- DBMS is often part of a larger software infrastructure
  - e.g., Database-backed web applications
- Integrating SQL with application code is a messy problem
  - Efficiency: Push computation “close” to data
  - Security: Validate input
  - Testing/Debugging

Next…

Normalization

$A \rightarrow BC$

$CD \rightarrow E$

$B \rightarrow D$

$E \rightarrow A$